Vector Search on Android — Finding What You Mean, Not What You Type

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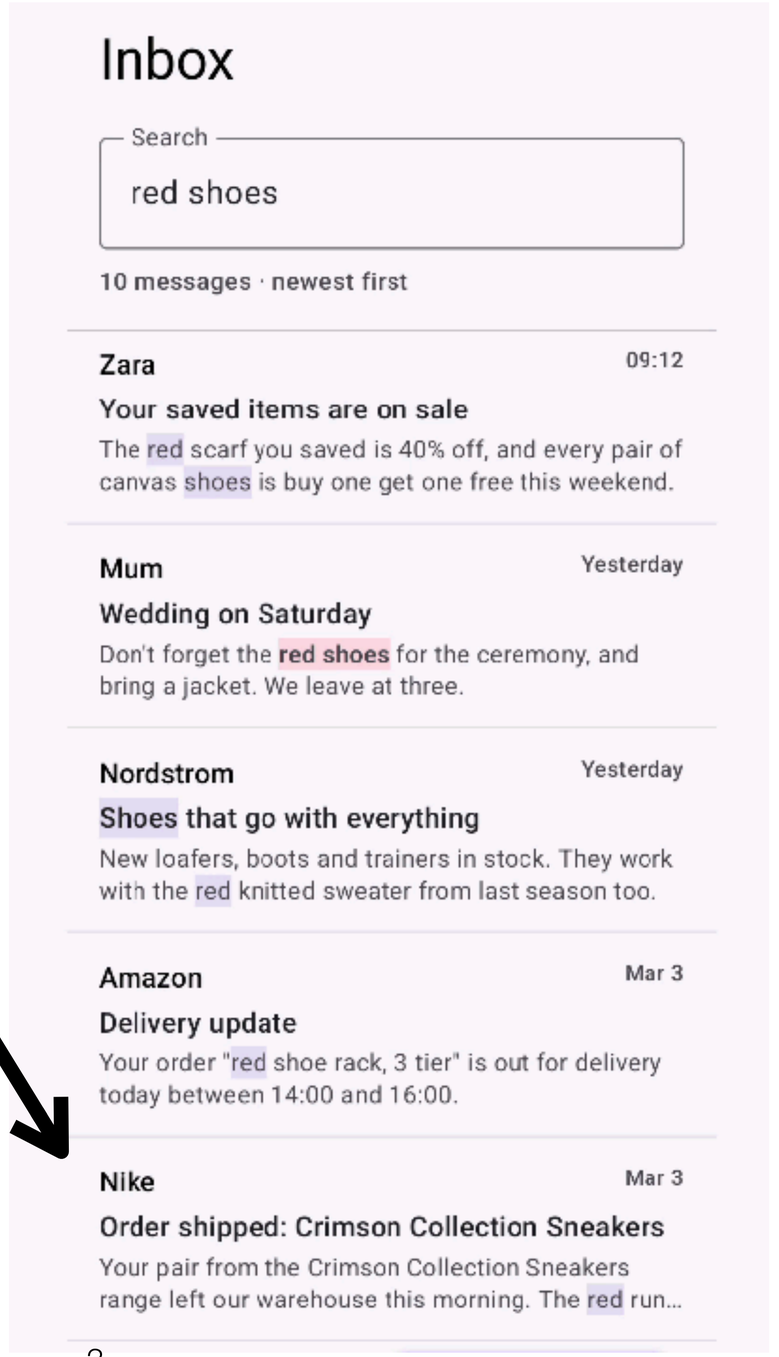
Layout

1. Problems with search
2. Vector Search
3. Components
4. Summary

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What is Vector Search?

Searching for “red shoes” in my email, gets me:



What is Vector Search?

What it should get me:



Inbox

Search

red shoes

10 messages · sorted by relevance

Nike Mar 3

Order shipped: Crimson Collection Sneakers

Your pair from the Crimson Collection Sneakers range left our warehouse this morning. The red run…

Mum Yesterday

Wedding on Saturday

Don't forget the **red shoes** for the ceremony, and bring a jacket. We leave at three.

ASOS Feb 27

Red shoes restocked

The red platform shoes are back in every size, plus 20% off boots with code SPRING.

The Rest Is History Feb 28

Episode 214: red shoes and power

Why did medieval kings wear **red shoes**? We ask a historian about footwear and status.

Riverside Running Club Mar 3

Sunday long run + club photo

The club photo is straight after the run. Bring your red running shoes if you own a pair.

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What makes it better than keyword or full text search?

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“Vector search” is a technique from semantic search.

What **components** do I need to do vector search?

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1. Embedding Model

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What components do I need to do vector search?

1. Embedding Model
2. Vector Store/Database

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What components do I need to do vector search?

1. Embedding Model
2. Vector Store/Database
3. Chunking

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Components: What is an **embedding model?**

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Plain machine learning model that has “mappings” from words to floating vectors.

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Those vectors are called “embeddings”.

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"cat" → [0.21, -0.44, 0.87, ...] (768 numbers)

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"cat" → [0.21, -0.44, 0.87, ...] (e.g. 768 numbers)

"kitten" → [0.19, -0.41, 0.85, ...] (close to "cat")

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"cat" → [0.21, -0.44, 0.87, ...] (e.g., 768 numbers)

"kitten" → [0.19, -0.41, 0.85, ...] (close to "cat")

"airplane" → [-0.62, 0.33, 0.11, ...] (far from "cat")

Components: What is *__exactly__* an embedding model?

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Components: What is *exactly* an embedding model?

Model trained on data to “learn” the relationships between words.

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Models supports single or multi-language.

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Models can be mono or multi-lingual.

The output size is constant, input has a maximum.

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Model trained on data to “learn” the relationships between words.

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The output size is constant, input has a maximum.

“Words” are represented as vectors of N size (dimensions).

Components: What are dimensions in an embedding model?

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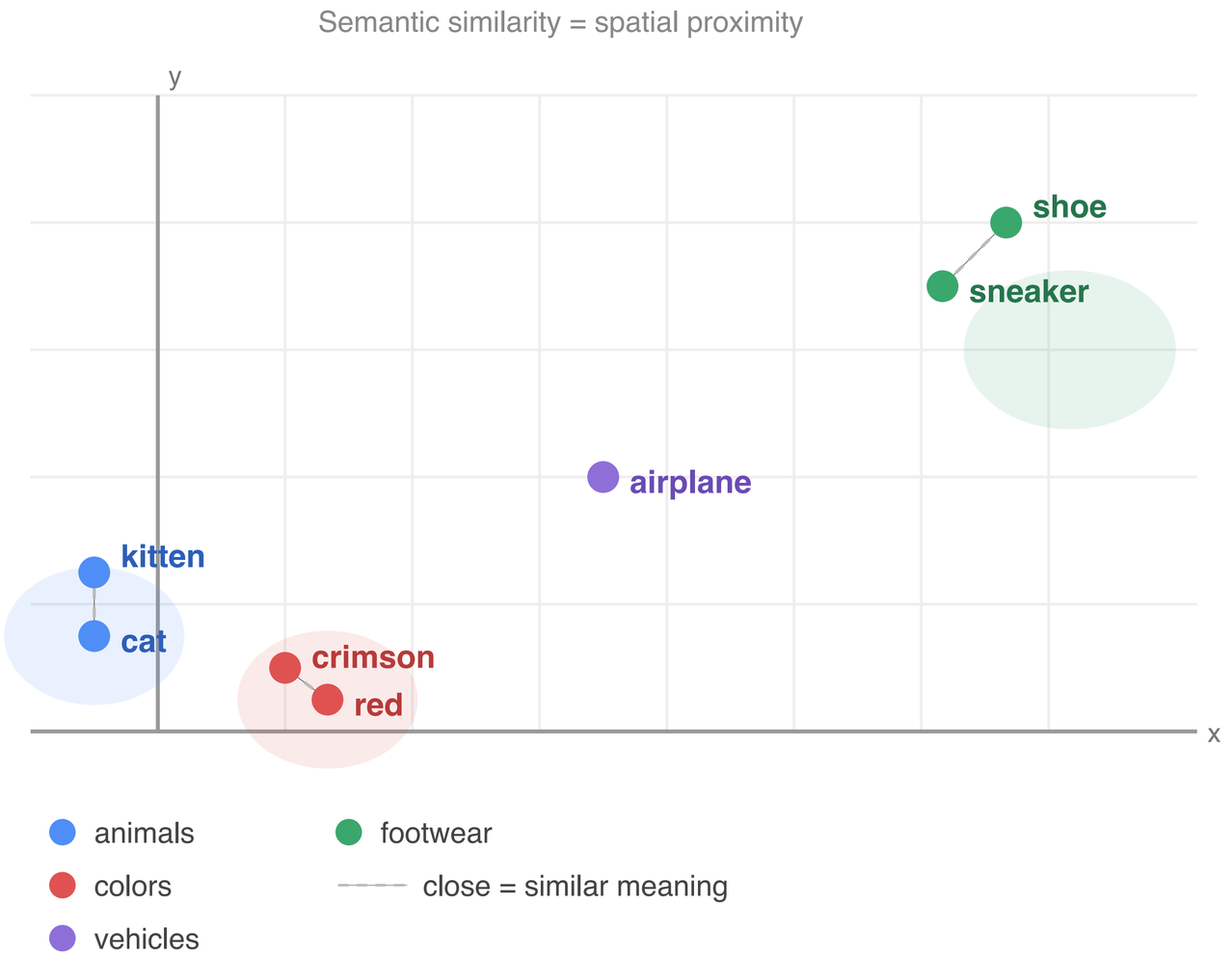
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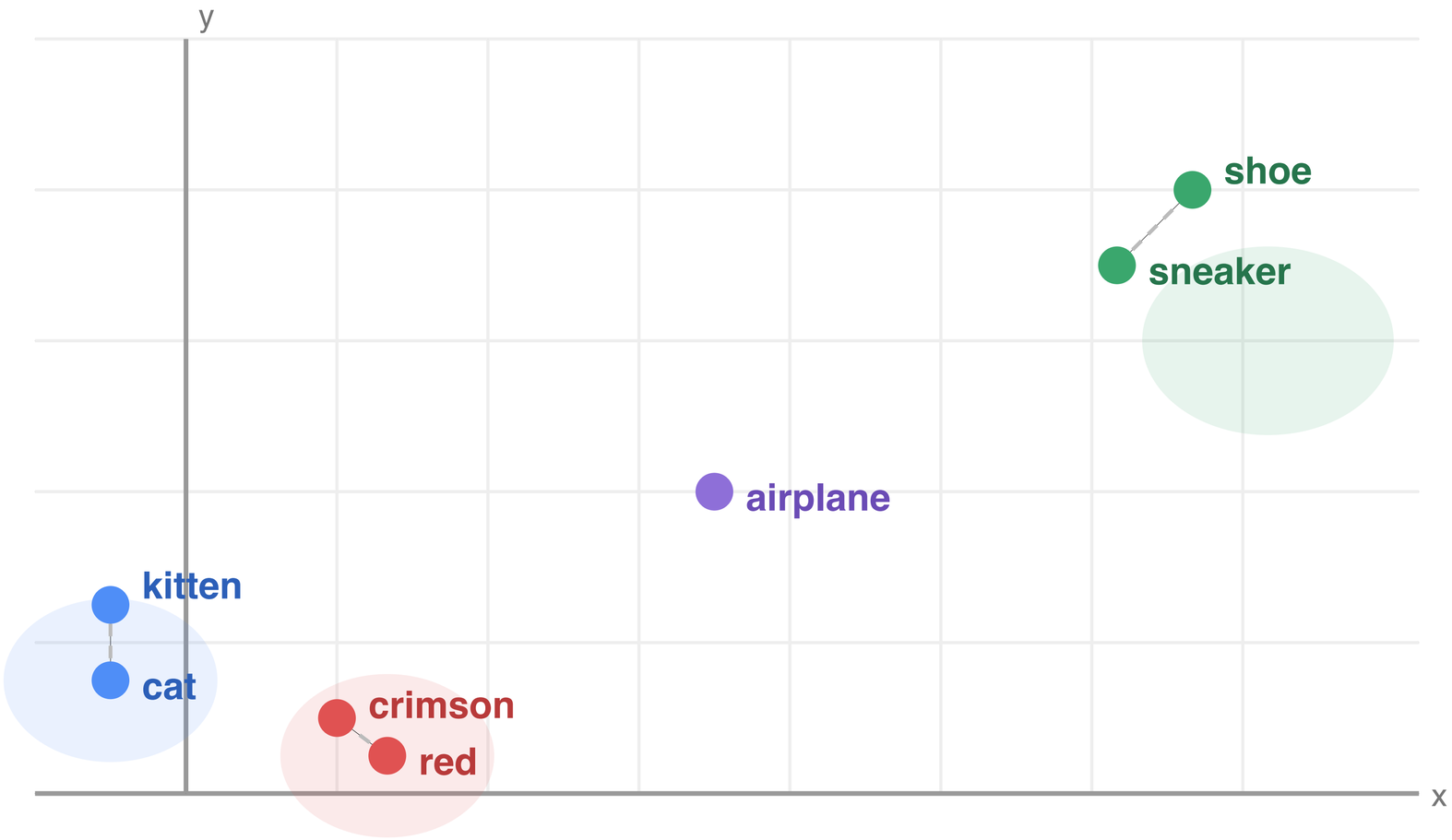
…

3072

Components: What are dimensions in an embedding model?



Semantic similarity = spatial proximity



animals
colors
vehicles
footwear
close = similar meaning

Fictional 2D projection — real models use 256–3072 dimensions

Components: What is used to **store the embedding vectors**?

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Vector Databases

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Vector Databases

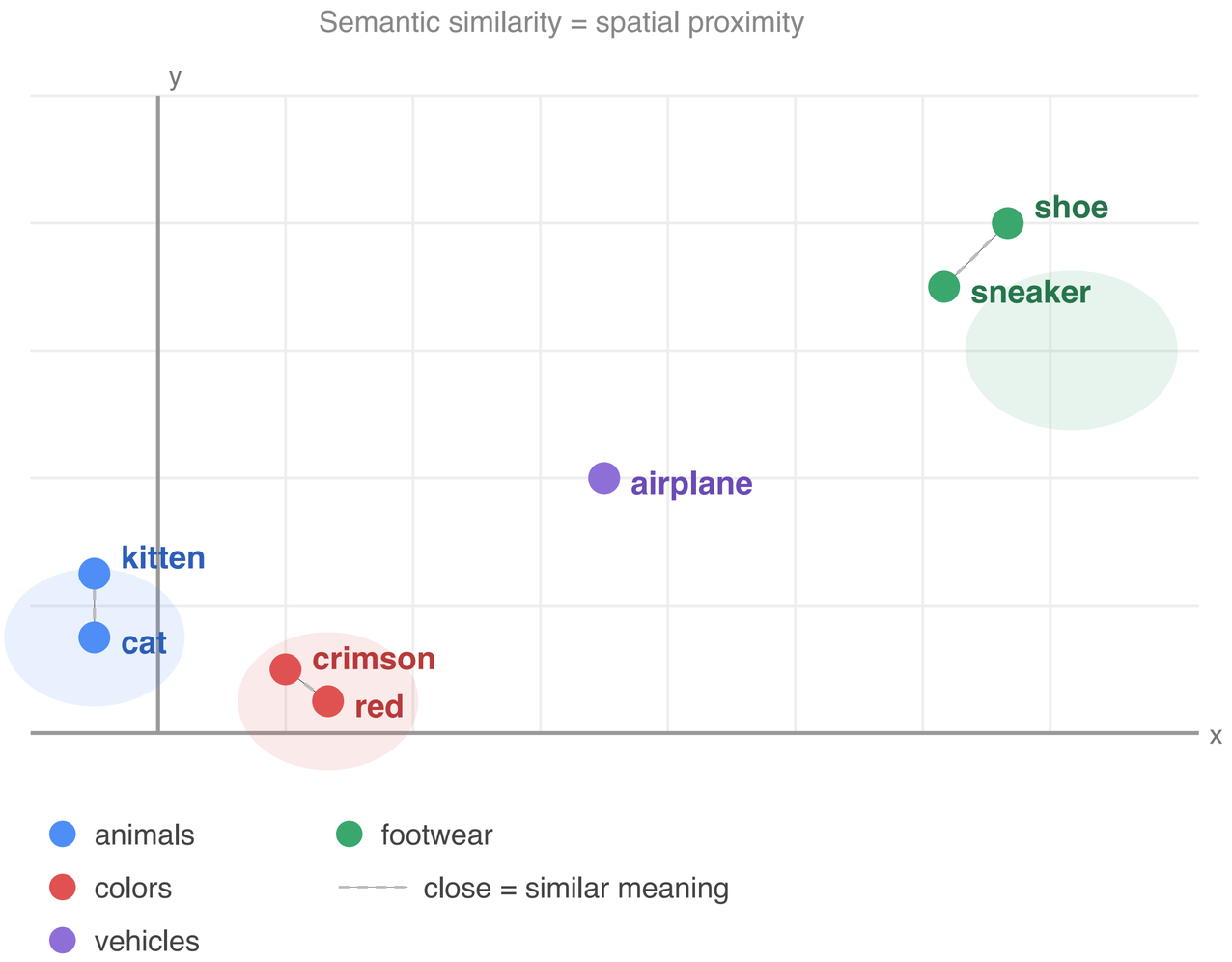
1. Efficiently store these vectors (float arrays)

2. Most come with ways to calculate the “similarity”

Components: What is similarity search?

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Fictional 2D projection — real models use 256–3072 dimensions

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Algorithms to calculate it:

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Your model will say which to use.

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Algorithms to calculate it:

1. **Cosine similarity** (great for text)
2. Euclidean Distance (great for spatial data)
3. Dot product (text for emphasis)

Your model will say which was trained on.

Components: What is **chunking**?

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Textual data needs chunking.

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Components: What is chunking?

It’s slicing.

Textual data needs chunking.

We as the builders know the data structure the best.

Components: What should be the size of the chunk?

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Model dependent!

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size * 4 = ~ total chars

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Try to retain complete sentences or slice on punctuation, because **of context**.

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We should also bring some of the **previous chunk to the next** one.

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We should not just slice blindly, by length, but look at the data.

Try to retain complete sentences or on punctuation, because of context.

We should also bring some of the previous chunk to the next one. **10-20%**

Components: What about chunking multiple fields?

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For example in an email with:

- Timestamp
- Sender
- Subject
- Body

Components: What about chunking multiple fields?

For example in an email with:

- Timestamp
- Sender
- Subject
- Body

We can just join all of it with spaces:

Timestamp Sender Subject Body

Components: What about chunking **images**?

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For images we don’t need to chunk.

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Images are a snapshot of fixed size.

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Components: What about chunking images?

For images we don’t need to chunk.

Images are a snapshot of fixed size.

A part of an image doesn’t give enough context.

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What does it **all** look like?

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What does it all look like?

There are 2 phases to implementing vector search:

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What does it all look like?

There are 2 phases to implementing vector search:

1. Embedding the stored data, to make it searchable

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What does it all look like?

There are 2 phases to implementing vector search:

1. Embedding the stored data, to make it searchable

2. Embedding the search query and performing the “similarity” search

When do I do the **embedding of the stored** data?

Whenever new searchable data is available.

If there is too much (e.g. first time) during low activity of your users.

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When do I do the embedding of the stored data?

Whenever new searchable data is available.

If there is too much (e.g. first time) during low activity of your users.

Note: don’t forget to save an ID to the original item.

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When do I do the **similarity search**?

Once the user tries to search for something.

You can then embed the search query, and calculate the similarity.

When do I do the similarity search?

Once the user tries to search for something.

You can then embed the search query, and then calculate the similarity.

Note: For a lot of vector data, filter by similarity score and limit to just X most similar.

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HuggingFace: search for embedding models.

There are also remote APIs that do the embedding for you.

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Model selection should depend on your ML runtime.

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Where can I find embedding models?

HuggingFace: search for embedding models.

There are also APIs that do the embedding for you.

Model selection should depend on your ML runtime.
LiteRT, ONNX, CoreML (iOS)

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Where can I apply vector search in my app?

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Where can I apply vector search in my app?

There are a few apps:

1. Email
2. Notes
3. Tutorials
4. Recipes
5. Image Gallery/Camera Roll

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Who can implement this?

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Who can implement this?

Anyone of you, after this talk.

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Summary

You need:

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You need:

1. Embedding Model - HuggingFace or Remote API

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Summary

You need:

1. Embedding Model - HuggingFace or Remote API
2. Vector Database - ObjectBox or sqlite-vec

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Summary

You need:

1. Embedding Model - HuggingFace or Remote API
2. Vector Database - ObjectBox or sqlite-vec
3. Chunking strategy

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Summary

What you should consider:

Summary

What you should consider:

a. Model Size

Summary

What you should consider:

a. Model Size

b. Multi-language support

Summary

What you should consider:

a. Model Size

b. Multi-language support

c. Different models = New full DB embedding on the data

Thank you, and that’s it!

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